

A large homology group can still be missing the required class

Unreduced Khovanov ranks over F_2 · red cells violate the ribbon injection inequality

K_1 required	1	1	1	1	2	1	1	1
J24227 — excluded	7	0	0	0	0	0	0	0
J25789 — excluded	0	4	5	6	5	4	3	1
J25533 — retained	2	1	3	4	5	6	5	4
	(-5, -5)	(0, -3)	(1, -1)	(2, 1)	(3, 3)	(4, 5)	(5, 7)	(6, 9)

Bigrading (homological, quantum)

J25789 and J25533 differ by one full band twist; both have knot Floer rank 173.

Retained means this necessary test passes. No K_0 - K_1 concordance or slice disk for their difference is known.